

1. Setup

Create a new notebook and add this as your very first code cell to install the official library.

```
# Install the modern Google GenAI SDK
!pip install -q -U google-genai
```

52.4/52.4 kB	2.6 MB/s	eta
0:00:00		
783.6/783.6 kB	21.9 MB/s	eta
0:00:00		
240.6/240.6 kB	24.2 MB/s	eta
0:00:00		

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
google-colab 1.0.0 requires google-auth==2.47.0, but you have google-auth 2.49.2 which is incompatible.

2. Authentication

```
from google import genai
from google.colab import userdata

# Securely fetch the API key
try:
    api_key = userdata.get('GEMINI_API_KEY')
    print("API Key successfully retrieved from Colab Secrets.")
except Exception as e:
    print("Error: Could not find 'GEMINI_API_KEY' in Colab Secrets.")

# Initialize the modern client
client = genai.Client(api_key=api_key)

API Key successfully retrieved from Colab Secrets.
```

3. API Request

```
from google.genai import types

# Define your prompt
prompt = "Briefly explain the physical mechanism of using a primary coil to transmit power to a secondary coil located inside an implanted neural stimulator chip."
```

```

print("Sending request to Gemini API...\n")

# Generate the response using the current 2.5 Flash model
response = client.models.generate_content(
    model='gemini-2.5-flash',
    contents=prompt,
    config=types.GenerateContentConfig(
        system_instruction="You are an expert engineering assistant
specializing in biomedical electronics.",
        temperature=0.7,
        max_output_tokens=1024 # Increased from 150 to give it plenty
of room to talk!
    )
)

print("--- API Response ---")
print(response.text)

```

Sending request to Gemini API...

--- API Response ---

The physical mechanism relies on **electromagnetic induction**, specifically **mutual induction** (or inductive coupling).

1. An external **primary coil** is driven by an alternating current (AC